HDMI-CEC and "Show the Camera on the TV" — Research Report

Date: 2026-10-26. Owner: CrystalHeeler. Scope: AnyCam add-on on a Raspberry Pi 4 under Home Assistant OS (HAOS), with go2rtc for streams.

Short answer

HDMI-CEC alone cannot do the job. HDMI Consumer Electronics Control (CEC) can wake a TV and switch it to an input. It cannot send a picture. The picture must come from the HDMI device that sends the CEC command. So CEC works only for one TV that has a cable to the Pi (or to a box that shows the camera).

For "any TV in the house", use the network, not the cable. The best fit is an overlay app on Android TV, Google TV or Fire TV. Examples are TvOverlay and the PiPup family. These apps put a live camera window over the current program. They do not stop the program. Google Cast is a second choice: it is full screen, but users report 10–30 s start delay (HA Community).

Recommended AnyCam design: AnyCam fires a "show on TV" request per camera, with a cooldown and quiet hours. A Home Assistant action does the TV work. AnyCam supplies the go2rtc stream URL. Details are in [Part 6](#6-design-suggestions-for-anycam).

Reddit: not reachable. The fetch tool refused `old.reddit.com`, and the web search returned no Reddit pages.

Method note: pages on community.home-assistant.io and IP Cam Talk returned HTTP 403 to the fetch tool. Claims from those pages come from search-result excerpts of the page text.

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1. What HDMI-CEC is

1.1 The bus

- CEC is one shared wire in the HDMI cable. It is separate from the video signals. It links all HDMI ports of a device, and it can work when the device is in standby (Wikipedia).
- The bus is slow: about 417 bit/s (Wikipedia).
- Up to 15 devices can share one bus (Wikipedia).
- The CEC wire is mandatory in HDMI. CEC features in a product are optional (Wikipedia).

1.2 Addresses

- **Logical address (4 bits).** Each frame header holds a 4-bit source and a 4-bit destination. Address 15 (0xF) is the broadcast address (Wikipedia, Pulse-Eight). The TV is logical address 0 (cec-ctl man page).
- **Claiming an address.** A device "pings" an address. If no device answers, it takes that address (Wikipedia).
- **Physical address.** This is the device's place in the HDMI tree, for example `1.0.0.0` for TV input 1. The `--active-source` command carries it (cec-ctl man page).

1.3 Opcodes that matter here

Values from Pulse-Eight:

Opcode	Name	What it does
`0x04`	Image View On	Wakes the TV and switches it to the sender's input.
`0x0D`	Text View On	Same as Image View On, and also clears TV menus.
`0x82`	Active Source	Broadcast: "I am now the source. Show my input."
`0x86`	Set Stream Path	The TV asks the device at a physical address to become the source.
`0x36`	Standby	Puts one device, or all devices on broadcast, into standby.
`0x44`	User Control Pressed	Sends a remote-control key press.
`0x80`	Routing Change	An HDMI switch reports a route change.

To show the Pi on the TV, the Pi sends **Image View On** to address 0, then broadcasts **Active Source** with its physical address (cec-ctl man page).

1.4 Brand names

Each brand sells CEC under its own name (Wikipedia):

- Samsung: Anynet+
- LG: SimpLink
- Sony: BRAVIA Sync, BRAVIA Link, Control for HDMI
- Panasonic: VIERA Link, HDAVI Control, EZ-Sync
- Philips: EasyLink
- Sharp: Aquos Link
- Toshiba: CE-Link, Regza Link
- Roku: 1-Touch Play
- Hisense, Vizio: CEC
- Onkyo: RIHD; Pioneer: Kuro Link; Mitsubishi: NetCommand for HDMI

1.5 Reliability between brands

- Manufacturers use different CEC implementations. They are not all compatible (Audioholics forum).
- Power on/off usually works across brands. Menu and input control fail more often (GTPlanet forum).
- Receivers sometimes lose the device list. The fix is to turn HDMI Control off and on (Audioholics forum).
- One plugin author writes that a command set that works on one TV may not work on another. Active Source can make a two-input TV flip to the wrong input (homebridge-cec-tv-control).
- A Pi can steal the input 1–2 s after the TV turns on, even with `hdmi_ignore_cec_init=1` (Plex forum).

Effect on AnyCam: expect brand-by-brand testing. "It works on my TV" does not carry to other TVs.

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2. Raspberry Pi 4 and CEC

2.1 Ports and device nodes

- The Pi 4 has two HDMI controllers. The kernel driver gives one CEC device per controller: `/dev/cec0` and `/dev/cec1` (Linux kernel mailing list).
- Use **HDMI0**, the port next to the USB-C power socket. Users report CEC works only there (Arch Linux ARM forum). Older libcec builds open only `/dev/cec0`, so only HDMI0 works with them (LibreELEC forum).
- Some micro-HDMI adapters do not carry the CEC pin. One user fixed CEC with a one-piece micro-HDMI-to-HDMI cable (LibreELEC forum). A kernel developer had the same adapter fault (Linux kernel mailing list).

2.2 Kernel CEC framework vs libcec

- **Kernel CEC framework.** The driver creates `/dev/cecX`. User programs talk to it through ioctl calls (kernel.org).
- **v4l-utils tools.** `cec-ctl` sends, receives and monitors CEC messages. `cec-compliance` tests a remote device. `cec-follower` emulates a device (kernel.org).
- **libcec.** This is the Pulse-Eight library, with the `cec-client` tool. It supports Pulse-Eight USB adapters and the Raspberry Pi (libcec on GitHub).
- Since Home Assistant 2021.7.0, its CEC libraries support only CEC devices in the Linux kernel (community Pi-CEC add-on page).

2.3 Commands: wake the TV and switch to the Pi

With `cec-ctl` (cec-ctl man page):

```
# 1. Register the Pi as a playback device (once per boot)
cec-ctl -d /dev/cec0 --playback

# 2. Wake the TV and ask it to show this input
cec-ctl -d /dev/cec0 --to 0 --image-view-on

# 3. Claim the input (use the physical address that cec-ctl reports)
cec-ctl -d /dev/cec0 --to 0 --active-source phys-addr=1.0.0.0

# Later: put the TV in standby
cec-ctl -d /dev/cec0 --to 0 --standby
```

With `cec-client`: `on 0` wakes the TV, `as` makes the Pi the active source, `is` marks it inactive (Pi My Life Up).

2.4 Known Pi problems

- A user on kernel 6.18 found that the first power-on after boot failed. Adding `--from 14` to `cec-ctl` fixed it (Arch Linux ARM forum).
- A HAOS user on a Pi 4 (HAOS 18.1, HA 2026.7.2) gets "autodetect FAILED" from the CEC Scanner app. The thread shows no fix (HA Community).

- libcec suggests `hdmi_force_hotplug=1` in `/boot/config.txt` when the Pi does not detect the TV (libcec on GitHub).

2.5 The hard limit: the cable

CEC reaches only devices on the same HDMI chain. The Pi must connect to the TV by HDMI, directly or through an audio/video receiver (AVR). One Pi with two ports can reach two TVs at most, and in practice only HDMI0 has CEC. A TV in another room needs its own HDMI source.

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3. Home Assistant and CEC

3.1 The `hdmi_cec` integration

- Actions: `power_on`, `select_device` (make a device the active source), `standby`, `send_command` (raw CEC), `volume`, `update` (HA docs).
- It creates `switch` entities by default, or `media_player` entities if you choose (HA docs).
- It is marked as a **legacy integration** and uses YAML setup (HA docs).
- On HAOS, the docs say to test with the CEC Scanner app, and to stop that app from starting on boot, because it blocks the integration (HA docs).
- It can also connect over TCP to a remote CEC server through the `host` option (HA docs).

3.2 Can an add-on open `/dev/cec0`?

Yes, in principle. The add-on `config.yaml` has these hardware options (HA developer docs):

Option	Effect
`devices: [/dev/cec0]`	Maps one named device into the container. This is the narrowest choice.
`video: true`	"Give access to all video devices." The docs do not say if this includes `/dev/cec*`.
`full_access: true`	All hardware. Works only with protection mode off. Lowers the security rating.
`homeassistant_api: true`	Lets the add-on call the Core API at `http://supervisor/core/api` as an administrator.

Proof from other projects:

- cec2mqtt runs in Docker with `/dev/cec0` mapped in. It reports power state and active source over MQTT with HA discovery. It is still in development.
- A community Pi-CEC add-on runs a CEC server on the Pi. The `hdmi_cec` integration connects to it over TCP.

Not tested here: whether `/dev/cec0` exists on the current HAOS Pi 4 kernel, and whether `video: true` maps it. AnyCam must test both on the hardware.

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4. Can CEC make a TV show a camera?

No, not by itself. CEC sends commands only. "Image View On" and "Active Source" tell the TV which HDMI input to show (Pulse-Eight). The pixels come from the device on that input.

So the CEC path needs two parts on the same box:

1. A CEC command to switch the TV to the box's input.
2. A program that draws the camera on the box's HDMI output.

4.1 Can the HAOS Pi draw video on its HDMI port?

- Stock HAOS is headless. It shows only the console on HDMI (HA Community).
- The community **HAOSKiosk** add-on starts X, Openbox and the Luakit browser, and shows a dashboard on the HAOS screen. One user on an Intel NUC saw Luakit crash. Another saw the screen go black after about 5 minutes (HA Community).
- So an add-on **can** drive the HDMI port. A browser page with a go2rtc player is one way. A video player such as mpv is another way. No source here shows mpv on HAOS. Both need display-device access and testing.

4.2 Practical verdict for CEC

- It fits one TV next to the Pi, cabled to HDMI0.
- The TV must be on the Pi's input to show anything. The TV leaves the current program. This is full screen, not picture-in-picture (PiP).
- Return to the old input is not reliable. "Inactive Source" asks the TV to choose, and each brand acts differently (homebridge-cec-tv-control).
- Most TVs no longer offer PiP from a second HDMI input. An IP Cam Talk member put it as "PiP is basically dead" (IP Cam Talk).

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5. Alternatives that need no HDMI cable

5.1 Android TV / Google TV / Fire TV — overlay apps (best fit)

TvOverlay

- Android TV app with a REST API and MQTT. Its `video` field takes a URL and "supports RTSP, HLS, DASH, and SmoothStreaming" (TvOverlay on GitHub).
- The repo has sample Home Assistant YAML. With MQTT, HA creates a TvOverlay device (TvOverlay on GitHub).
- An openHAB binding author reports RTSP camera video on the TV with under 1 s lag (openHAB Community).
- Needs "Draw over other apps". On some devices you grant it through ADB. The Play Store listing was suspended once (TvOverlay on GitHub).

PiPup and its fork

- PiPup listens on port 7979 for `POST /notify`. Media can be `image`, `video` or `web`, with a size and one of 5 screen positions (PiPup on GitHub).
- Sideloaded installs need `adb shell appops set nl.rogro82.pipup SYSTEM_ALERT_WINDOW allow` (PiPup on GitHub).
- The original repo is unmaintained. A community HA integration adds a real `pipup` action, through the ADB integration (HA Community).
- A maintained fork adds popups with no time limit (`duration: 0`), buttons, TTS and screen on/off. It was tested on Fire OS, Google TV and Android TV. Popup video plays silent by default, because audio froze live TV on some devices (HA Community).
- Users report mixed video results: one needed MJPEG, and one saw an RTSP stream crash the app (Homey Community).

Notifications for Android TV / Fire TV (`nfandroidtv`)

- Built-in HA integration. Shows a still image (URL or file), with duration, position and transparency. No video (HA docs).
- The display options work only with the legacy `notify.<name>` action, not `notify.send_message` (HA docs).
- Use it as a fallback: a snapshot from go2rtc `/api/frame.jpeg` (go2rtc).

Android Debug Bridge (ADB) integration (`androidtv`)

- Controls Android TV and Fire TV over ADB. `media_player.select_source` launches an app by app ID. `androidtv.adb_command` sends keys or shell commands (HA docs).
- Only one ADB client can connect at a time. Wi-Fi drops can break the link (HA docs).
- Use: wake the TV, open a full-screen camera app, then return to the old app.

Fire TV notes

- Fire OS 8 has no on-screen way to grant the overlay permission. You grant it through ADB (Play Digital Signage).
- IP Cam Talk users pop a camera on Fire TV with TinyCam Pro and Tasker (IP Cam Talk).

Platform note. Android TV allows true system PiP for "smartHome" uses such as doorbells. It does not allow PiP for media playback (Android Developers). Overlay apps use a different window type, so this rule does not block them.

5.2 Google Cast (Chromecast, Google TV, Nest Hub)

- `camera.play_stream` sends a camera to a media player. Default format is HLS. It needs the `stream` integration (HA docs).
- `media_player.play_media` plays any HTTP(S) URL. Cast devices use Google DNS, so `homeassistant.local` names fail; use an IP address (HA docs).
- `cast.show_lovelace_view` shows a dashboard on the TV. It needs HTTPS access to HA (HA docs).
- Discovery needs mDNS on the same subnet (HA docs).

- Latency: users report 20–30 s to start (HA Community) and 10–30 s (HA Community). One tester saw HLS about 30 s behind live, and MJPEG about 0.5 s behind (openHAB Community).
- go2rtc rates HLS latency "bad", MSE "medium", WebRTC "best" (go2rtc).
- Casting replaces the current program. It is full screen.

5.3 Apple TV

- tvOS shows a PiP window over the current video when a HomeKit doorbell rings (AppleInsider). Apple's guide confirms live camera PiP and doorbell alerts (Apple Support).
- The HA HomeKit Bridge can expose a camera with `linked_doorbell_sensor` and `linked_motion_sensor` (HA docs).
- Not confirmed: that an HA-bridged camera gets the same PiP popup on Apple TV. The HA page does not say.

5.4 LG webOS

- The HA integration launches apps (`select_source`), sends commands (`webostv.command`) and shows text notifications (HA docs).
- The notification icon must be a local file. Many newer firmware versions ignore it (HA docs).
- Power-on needs Wake-on-LAN or CEC (HA docs).
- One community project shows doorbell snapshots as an overlay, but needs a rooted TV with SSH (HA Community).
- No source here shows live camera video over a program on webOS.

5.5 Samsung Tizen

- The HA integration sends keys and switches sources. It does not document URL or media playback (HA docs).
- A feature request for camera popups on the custom SamsungTV Smart integration has no finished build (GitHub issue).
- Samsung's own route: Ring Doorbell Pro video on 2022+ TVs through SmartThings (Samsung Support). That is a closed path, not open to AnyCam.

5.6 Kodi

- Kodi notifications accept an image URL as the icon, with a display time (HA docs).
- Kodi can wake the TV over CEC through the `script.json-cec` add-on (HA docs).
- A Kodi box on an HDMI port is a good "CEC + picture" box: it wakes the TV and plays the stream itself.

5.7 Roku

- `media_player.play_media` plays URLs, including `hls` and `mp4`. `camera.play_stream` sends HLS camera streams to a Roku (HA docs).
- Newer Roku OS versions may need a helper app for URL playback (HA docs).

- Full screen. HLS latency applies.

5.8 Comparison table

Approach	Works from an HAOS add-on?	Needs HDMI cable?	Interrupts viewing?	Full screen or PiP	Latency (reported)	Effort
CEC from the Pi + Pi draws video	Partly: CEC via `devices`; drawing video needs a kiosk-type add-on	Yes, Pi HDMI0 to TV	Yes, switches input	Full screen	CEC bus is fast; TV wake time not measured	High; brand testing
Android/Google/ Fire TV overlay (TvOverlay, PiPup fork)	Yes, HTTP or HA action	No	No, overlay on top	PiP window	Under 1 s reported for RTSP (openHAB)	Medium: sideload, ADB permission
`nfandroidtv` snapshot	Yes, HA action	No	No	Small image popup	Snapshot only	Low
ADB: open a camera app full screen	Yes, HA action	No	Yes, replaces app	Full screen	App start time; not measured	Medium
Google Cast (`camera.play_stream` / `play_media`)	Yes, HA action	No	Yes, replaces program	Full screen	10–30 s start (HA Community)	Low
Apple TV via HomeKit Bridge	Yes, HA config	No	No	PiP (doorbell)	Not measured	Medium; PiP for bridged cams unconfirmed
LG webOS	Partly: text and app launch	No	Text popup only	No live video	n/a	Low; no live video
Samsung Tizen	No open path	No	n/a	n/a	n/a	Not available
Kodi box	Yes, HA action	Box to TV	Yes	Full screen or notification image	Not measured	Medium
Roku	Yes, HA action	No	Yes	Full screen	HLS delay	Low

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6. Design suggestions for AnyCam

6.1 Split the job

- **AnyCam decides** when a camera should go to the TV: motion, cooldown, quiet hours.
- **Home Assistant does** the TV action. HA already has the TV integrations in Part 5.
- AnyCam calls HA through the Supervisor proxy: `POST` `http://supervisor/core/api/services/<domain>/<service>`, with `homeassistant_api: true` in `config.yaml` (HA developer docs).
- Simplest first step: AnyCam fires an HA event (for example `anycam_show_on_tv`) with the camera name and stream URLs. The user writes one automation per TV. AnyCam does not need to know TV brands.

6.2 Per-camera settings

- `show_on_tv`: on/off.
- `tv_targets`: list of HA entities (media players, notify services) or overlay app URLs.
- `tv_mode`: `overlay` (default), `fullscreen`, or `snapshot`.
- `tv_duration_s`: how long to show (default 30).
- `tv_cooldown_s`: no repeat for this camera and TV within this time (default 300).
- `tv_quiet_hours`: start and end time; no TV action inside the window.
- `tv_min_confidence`: use the object engine result, so trees and shadows do not hijack the TV.

6.3 Stream URLs for TVs

- Give the TV a go2rtc URL on the LAN, not an HA cloud URL. go2rtc offers RTSP on port 8554, HLS at `/api/stream.m3u8?src=<cam>`, MP4 at `/api/stream.mp4?src=<cam>`, and snapshots at `/api/frame.jpeg` (go2rtc).
- Prefer RTSP for overlay apps (low delay). Use HLS or MP4 only for Cast and Roku.
- Use an IP address, not a `.local` name. Cast devices cannot resolve local names (HA docs).
- Prefer an H.264 substream. H.265 support differs by device. go2rtc lists H.265 as missing in Firefox and limited in other players (go2rtc).
- Security: the go2rtc API on port 1984 has no password by default (WebRTC Camera project). Do not expose it outside the LAN.

6.4 Do not hijack the TV

- Cooldown per camera and per TV.
- One popup per TV at a time. A second camera replaces the first, or queues.
- Quiet hours.
- Overlay first, full screen only on request.
- A global "TV alerts off" switch as an HA entity.

6.5 Return to the previous content

- **Overlay apps:** the popup closes after `duration`. Nothing to restore (PiPup on GitHub).
- **ADB full screen:** read the current app ID first. After N seconds, launch that app again with `select_source` (HA docs).
- **Cast:** call `media_player.media_stop` after N seconds. The program that was playing before does not come back by itself.
- **CEC:** send "Inactive Source" or ask the user to press Back. The TV's reaction differs by brand (homebridge-cec-tv-control).

6.6 CEC path, if wanted

- Only for one TV cabled to the Pi on HDMI0.

- Add `devices: [/dev/cec0]` to `config.yaml`. Install `v4l-utils` in the image. Test that `cec-ctl -d /dev/cec0 --playback` sees the TV.
- Send `--image-view-on`, then `--active-source`. Use `--from` as in the kernel 6.18 report (Arch Linux ARM forum).
- The Pi must also draw the camera full screen on HDMI. That is a separate, larger feature (kiosk browser or player). Treat it as a later step.

6.7 Suggested order of work

1. HA event + example automations (overlay app, Cast, `nfandroidtv` snapshot). Low risk.
2. Cooldown, quiet hours, confidence gate in AnyCam.
3. Direct calls to TvOverlay / PiPup HTTP APIs from AnyCam, with no HA automation needed.
4. CEC on the Pi, only if a TV is cabled to it.

Not tested

- No TV, Pi or HAOS hardware was tested for this report. All findings come from documents and user reports.
- Not checked: `/dev/cec0` on the current HAOS Pi 4 kernel; `video: true` mapping of CEC devices; mpv or a browser player on HAOS; Apple TV PiP for HA-bridged cameras.
- Latency numbers are user reports, not measurements.

Sources

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3. kernel.org — CEC introduction
4. cec-ctl man page (ManKier)
5. libcec on GitHub
6. Linux kernel mailing list — vc4 HDMI CEC for BCM2711
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9. Arch Linux ARM forum — Getting CEC working on RPi 4
10. Arch Linux ARM forum — kernel 6.18 CEC issue
11. Pi My Life Up — Using HDMI-CEC on a Raspberry Pi
12. Plex forum — RasPlex always switches TV input
13. npm — homebridge-cec-tv-control

14 Audioholics forum — TV on/off sync issue

15 GTPlanet forum — HDMI control between brands

16 Home Assistant — HDMI-CEC integration

17 HA Community — Issues with CEC

18 rfc1149.net — Home Assistant Pi-CEC add-on

19 cec2mqtt on GitHub

20 Home Assistant developer docs — App (add-on) configuration

21 HA Community — HAOSKiosk add-on 0.9.8

22 HA Community — HA dashboard via HDMI on TV

23 Home Assistant — camera.play_stream action

24 Home Assistant — Stream integration

25 Home Assistant — Google Cast integration

26 HA Community — Playing camera stream to Chromecast

27 HA Community — camera.play_stream lag issues

28 openHAB Community — Chromecast + HLS video stream

29 go2rtc on GitHub

30 WebRTC Camera on GitHub

31 Home Assistant — Android Debug Bridge integration

32 Home Assistant — Notifications for Android TV / Fire TV

33 PiPup on GitHub

34 HA Community — PipUp notification service

35 HA Community — PiPup fork + Home Assistant integration

36 Homey Community — PiPup app thread

37 TvOverlay on GitHub

38 openHAB Community — Android Notifications binding with TV Overlay

39 Android Developers — Multitasking on TV

40 Play Digital Signage — Fire TV launch on startup (Fire OS 8 overlay permission)

41 IP Cam Talk — Blue Iris PiP on TV when motion detected

42 IP Cam Talk — Show camera on Fire device on motion

43 AppleInsider — tvOS 14 HomeKit video PiP

44 Apple Support — Cameras in the HomeKit tab on Apple TV

45 Home Assistant — HomeKit Bridge integration

46 Home Assistant — LG webOS TV integration

47 HA Community — Doorbell snapshots as an overlay on LG webOS

48 Home Assistant — Samsung Smart TV integration

49 GitHub — SamsungTV Smart issue 106

50 Samsung Support — Ring Doorbell Pro on Samsung TV

51 Home Assistant — Kodi integration

52 Home Assistant — Roku integration

Reddit (reddit.com) was tried one time and was not reachable.